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JAPHET BITRUS

THE FEDERAL POLYTECHNIC MUBI ADAMAWA STATE LECTURE NOTE

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COURSE CODE	EEC 115
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COURSE LECTURE	A.S.KITAKO/AMINU MOH'D

1.1 ATOM

Defination: An atom is the smallest indivisible particle which all matter is made of up.

1.2 STRUCTURE AND COMPOSITION OF AN ATOM

According to the electron theory of matter, an atom is made up of tiny particles. The central part of the atom is known as the nucleus. The nucleus of the atom consists of particles called protons and neutrons held together in a compact form by a binding energy. The outer portion of the atom is made up of orbiting particles called electrons. Electrons normally move about the centre of an atom in paths which are referred to as shells orbits. Each of these shells can contain only a certain maximum number of electrons. The central part of the atom shows its nucleus. We should note that protons are positively charged particles while electrons are negatively charged particles, since opposite charges attract, the negatively charged electrons are attracted to the positively charged protons in the nucleus. Consequently, this electric force of attraction holds the electrons in their orbit. The number of orbiting electrons normally equals the number of protons in the nucleus. For that reason an atom is said to be neutrally charged.

1.3 CONDUCTORS, INSULATORS AND SEMICONDUCTOR

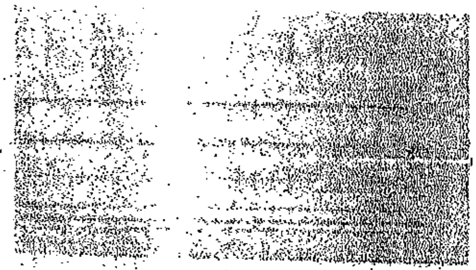
Conductor

In a conductor, electric current can flow freely, in an insulator it cannot. Metals such as copper typify conductors, while most non-metallic solids are said to be good insulators, having extremely high resistance to the flow of charge through them. "Conductor" implies that the outer electrons of the atoms are loosely bound and free to move through the material. Most atoms hold on to their electrons [illegible]tly and are insulators. In copper, the valence electrons are essentially [illegible] and strongly repel each other. Any external influence which moves one [illegible] them will cause a repulsion of other electrons which propagates, "domino fashion" th[illegible]gh the conductor.

Types of [illegible]uctors and their properties

Many types [illegible] materials can conduct electricity. Take these materials listed here and put the[illegible] in order from lightest to heaviest with respect to their densities, that is, weig[illegible] per unit volume:

Alminium, [illegible]pper, Gold, Iron, Lead, Mercury, Silver, Water, Zinc



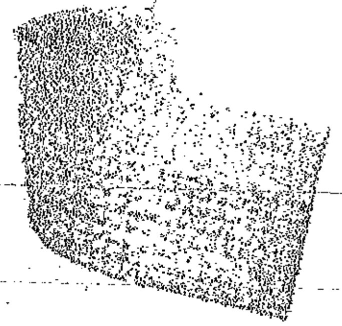
aluminum

copper

Insulator

An Insulator is a material or object which resists the flow of heat (thermal insulators) or electric charge (electrical insulators).

The term *insulator* has the same meaning as the term *dielectric*, but the two terms are used in different contexts. The opposite of insulators are conductors and semiconductors, which permit the flow of charge. Semiconductors are strictly speaking also insulators, since they prevent the flow of electric charge at low temperatures, unless doped with atoms that release extra charges to carry the current). However, some materials (such as silicon dioxide) are very nearly perfect electrical insulators, which allow flash memory technology. A much larger class of materials, (for example rubber and many plastics) are "good enough" insulators to be used for home and office wiring (into the hundreds of volts) without noticeable loss of safety or efficiency.



Rubber

wood

Semiconductor

A **semiconductor** is a material with an conductance that is intermediate between that of an insulator and a conductor.

A semiconductor behaves as an insulator at very low temperature, and appreciable conductance at room temperature conductor by the fact that, at absolute zero filled in a semiconductor, but only partially filled in a conducto semiconductor and an insulator is slightly more arbitrary. A semiconductor has a which is small enough such that its electrons at room temperature, whilst an insulator has a band gap which is too wide for there to be appreciable thermal electrons in its conduction band at room temperature.

Band structure of a semiconductor Rubber

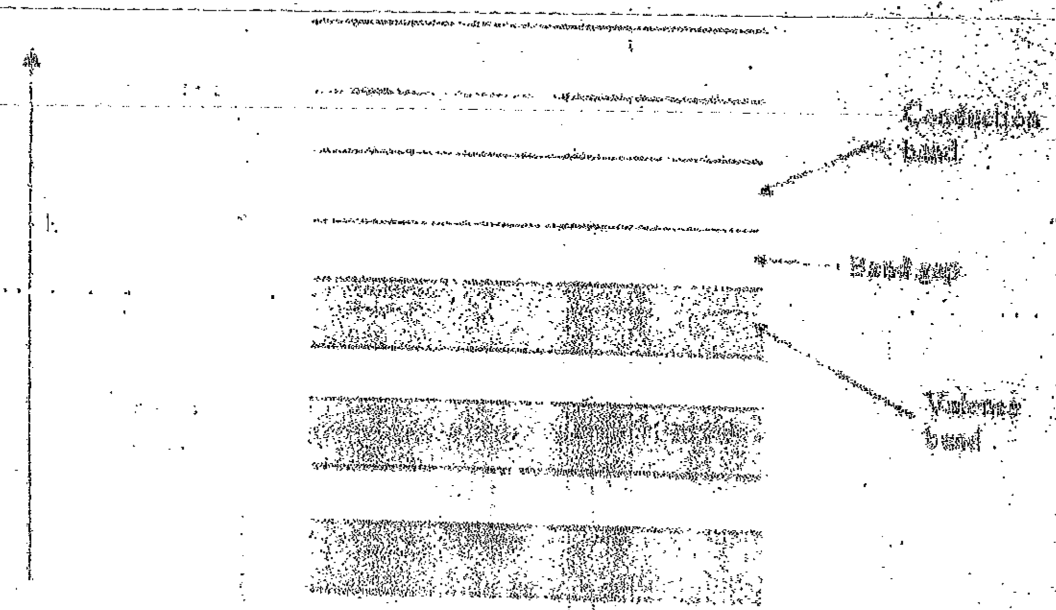


Fig 1.1: Band structure of a semiconductor showing a full valence band and an empty conduction band. The Fermi level lies within the forbidden band gap

In the parlance of solid-state physics, semiconductors (and insulators) are defined as solids in which at absolute zero (0 K), the uppermost band of *occupied* electron energy states, known as the valence band, is completely full. Or, to put it another way, the Fermi energy of the electrons lies within the forbidden band gap. The Fermi energy, or Fermi level can be thought of as the energy up to which available electron states are occupied at absolute zero.

At room temperatures, there is some smearing of the energy distribution of the electrons, such that a small, but not insignificant number have enough energy to cross the energy band gap into the conduction band. These electrons which have enough energy to be in the conduction band have broken free of the covalent bonds between neighbouring atoms in the solid, and are free to move around, and hence conduct charge. The covalent bonds from which these excited electrons have come now have missing electrons, or *holes* which are free to move around as well. (The holes themselves don't actually move, but a neighbouring electron can move to fill the hole, leaving a hole at the place it has just come from, and in this way the holes appear to move.)

It is an important distinction between conductors and semiconductors that, in semiconductors, movement of charge (current) is facilitated by both electrons and holes. Contrast this to a conductor where the Fermi level lies *within* the conduction band, such that the band is only half filled with electrons. In this case, only a small amount of energy is needed for the electrons to find other unoccupied states to move into, and hence for current to flow.

The ease with which electrons in a semiconductor can be excited from the valence band to the conduction band depends on the band gap between the bands, and it is the size of this energy bandgap that serves as an arbitrary dividing line between semiconductors and insulators. Materials with a bandgap energy of less than about 3 electron volts are generally considered semiconductors, while those with a greater bandgap energy are considered insulators.. The current-carrying electrons in the conduction band are known as "free electrons," although they are often simply called "electrons" if context allows this usage to be clear. The holes in the valence band behave very much like positively-charged counterparts of electrons, and they are usually treated as if they are real charged particles.

1.4 CONCEPT OF CURRENT AND ELECTRON FLOW

At any instant in time, the electrons in a conductor are in random motion. However, if a directional force e.g. electromotive force EMF from a battery is applied to the conductor as shown in Fig 2.1 below, then end A of the conductor is positive while end B is negative. This results in directed flow of electrons. The directional movement of free electrons is referred to as current flow and the conventional current flow is in the opposite direction of electron flow as can be seen in the diagram. Normally, conventional current flow is generally used.

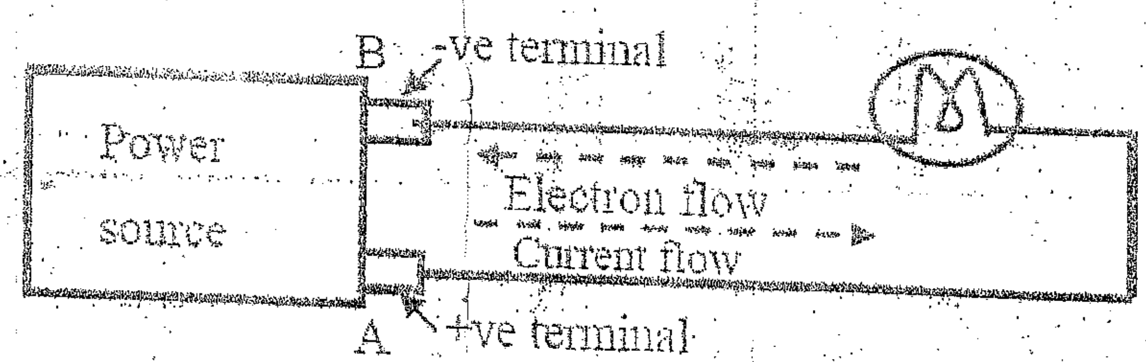


Fig 2.1

1.5 ELECTRIC CURRENT, POTENTIAL DIFFERENCE, ELECTROMOTIVE FORCE AND RESISTANCE

Electric Current Electric current is the rate of flow of electric charges round a circuit. It has a symbol I and is measured in amperes (A)

Potential Difference (P.d.) The potential difference between two points is defined as the number of joules of electrical energy transferred from one side of the points to the other with the passage of one coulomb from one point to another. It is measured in volts (V).

Electromotive Force (e.m.f) Electromotive force is the force that gives rise to electric current in a circuit; also it's the force that makes flow of electric current in a circuit. This force arises from many effects including chemical (e.g. battery cells) and magnetic, as (e.g. generator). The unit of electromotive force is the volt, symbol ε.

Resistance The resistance can be defined as an opposing force experience by the flow of charge through a material. The opposition is due to the collision between electrons and other atoms in the material, and it converts electrical energy into heat energy. The unit of resistance is Ohm (Ω), and has a symbol R.

Table 2.1. SI prefixes

PREFIX	SYMBOL	MULTIPLIER
Tera	T	10^{12}
Giga	G	10^{9}
Mega	M	10^{6}
Kilo	K	10^{3}
Hector	h	10^{2}
Deka	da	10^{1}
Deci	d	10^{-1}
Centi	c	10^{-2}
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}

2.1 DIRECT CURRENT

Direct current is that current which flows in only one direction at a time. The d.c. circuits normally contain a battery, which is the source, and this battery has two terminals namely: Negative and Positive terminals as shown in figure 3.1 below.

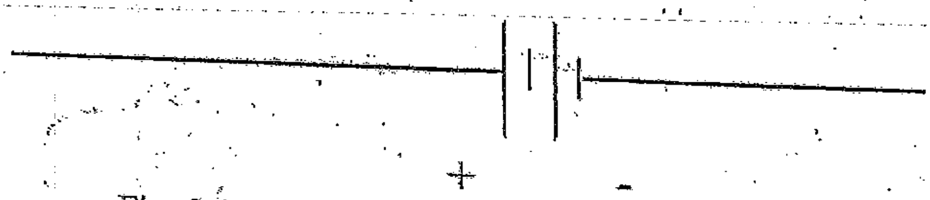


Fig. 3.1: symbol of a battery

2.2 ANALOGY BETWEEN CURRENT FLOW AND WATER FLOW

The three main concepts in electricity are voltage, current and resistance. Voltage or potential difference can be thought of as the driving force (although it is not really a "force") behind the electric current. In the water analogy, in figure (3.2), you can think of voltage as the pressure difference created by a pump that causes the water to flow through the pipe in the water system. Also you can think of the electrons flow in the electric circuit as equivalent to the flow of the water in the pipes that caused by the pump. Another important concept in electricity is resistance. It's Property of a material that opposes the flow of electrons through it. There is always a resistance flow between two points in a pipe. This is analogous to the resistance (R) in an electric circuit.

~~Three~~ main concept in electricity

i. Voltage – electrons flow in ~~a electricity~~ the electric circuit ~~current~~ as an equivalent to the flow of the water in the pipe caused by pump.

ii. Current – The rate of flow of electric charge round a circuit.

iii. Resistance – It is property of a material that opposes the flow of electron through it.

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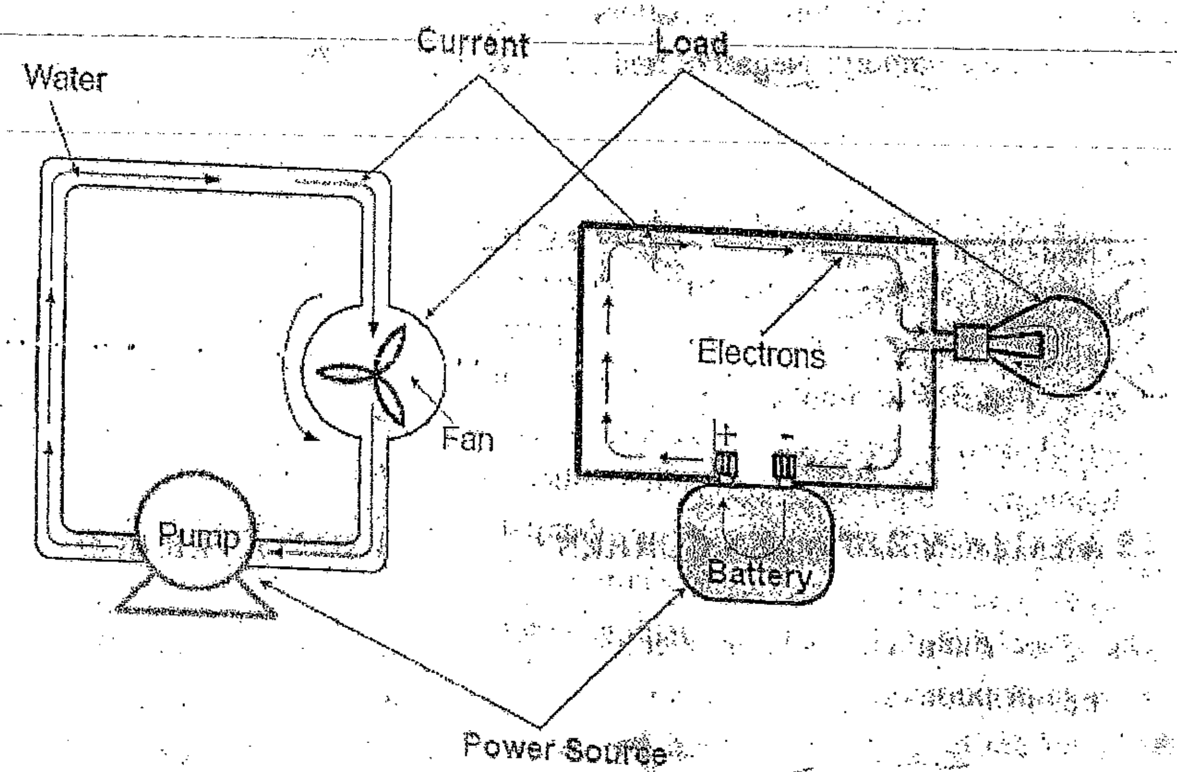


Figure (3.2) water analogy

2.3 BASIC D. C. CIRCUIT

A basic d.c. (direct current) circuit consist of an electrical connection of a d.c power source (e.m.f source), connecting wires and a load as shown in Fig 3.3. In this case, the load can be any electrical component (e.g. a resistor) or a network of resistors. The d.c. power sources such as dry cells (such as used in flash lights), car batteries and laboratory d.c power supply unit. The symbol of a d.c source is as shown in Fig. 3.3. It is represented by two vertical lines, with the longer line marked (+) and the shorter line marked (-).

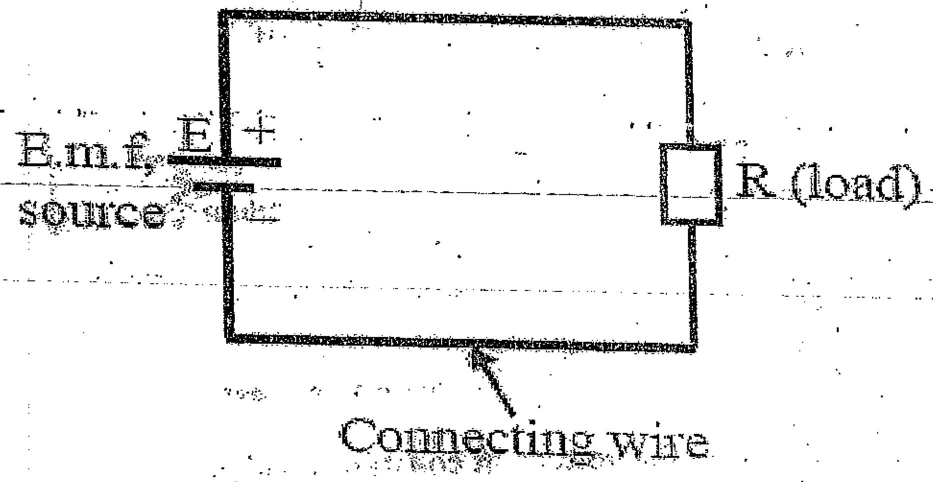


Fig 3.3

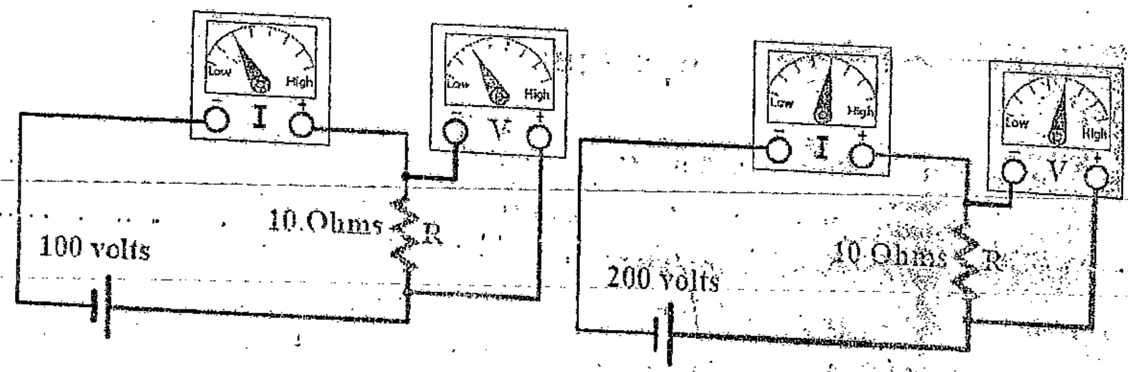
2.4 OHM'S LAW

In 1826 George Simon Ohm found that the current, voltage, and resistance are related in a specific way. Ohm expressed this relationship with a formula that is known today as Ohm's law. In this chapter, you will learn Ohm's law and how to used it in solving circuit problems. Electric circuit can be of two basic forms: series and parallel. In this chapter, series, and parallel circuits are studied. you will also see how Ohm's law is used in series, and parallel circuits

Ohm's law is the most important mathematical relationship between voltage, current and resistance in electricity. Ohm's Law is used in three forms depending on which quantity voltage, current or resistance you need to determine. In this section, you will learn each of these forms

2.4.1 Explanation of Ohm's Law

In the electric circuit, if the voltage across constant resistor value is increase, the current through the resistor will also increase; and, if the voltage is decrease, the current will decrease. For example, if voltage is double, the current will doubled. If the voltage is halved, the current will also be halved. This relationship is illustrated in the figure (4.1), with used voltage and current meter.

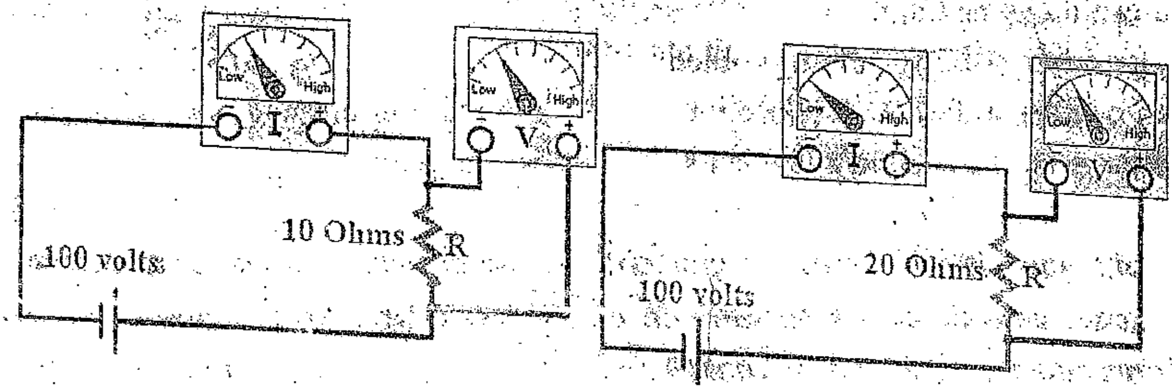


(a) Decrease V, decrease I

(b) Increase V, increase I

Figure (4.1): Effect of changing the voltage at constant resistance

Ohm's law also stated that if the voltage is kept constant, less resistance results in more current, and, also, more resistance results in less current. For example, if resistance is halved, the current will doubled. If the resistance is doubled, the current is halved. This relationship is illustrated by the meter indications in the figure (4.2). Where the resistance is increase and the voltage is constant.



(a) Decrease R, increase I

(b) Increase R, decrease I

Figure (4.2): Effect of changing the resistance at constant voltage

From previous illustrated,

Ohm's law states that, current and voltage are linearly proportional at constant resistance; current and resistance are inversely related at constant voltage.

According to this law the following three equivalent formula is derived:

I V∝ ∝R

Formula for current

$$I = \frac{V}{R} \quad A \qquad \text{Equation (4.1)}$$

This form of Ohm's low is used to determine current if voltage and resistance values are known

Formula for voltage

$$V = I\,R \quad V \qquad \text{Equation (4.2)}$$

This form of Ohm's low is used to determine voltage if current and resistance values are known.

Formula for resistance

$$R = \frac{V}{I} \quad \Omega \qquad \text{Equation (4.3)}$$

This form of Ohm's low is used to determine resistance if voltage and current values are known

Remember the following three rules:

Ohm's Law can only give the correct answer when the correct values are used.

Current is always expressed in Amperes or Amps (A)

Voltage is always expressed in Volts (V)

Resistance is always expressed in Ohms (Ω)

$$\frac{I}{1} \times \frac{V}{R} \qquad \frac{V}{I} = \frac{RI}{I} \quad R = \frac{V}{I}$$

$V = IR$

2.4.3 The Relationship of Current, Voltage and Resistance

Ohm's law describes how current is related to voltage and resistance. current and voltage are linearly proportional at constant resistance; current and resistance are inversely related at constant voltage.

2.4.3.1 The Linear Relationship of Current and Voltage

Current and voltage linearly proportional; that is, if one is increased by a certain percentage, the other will increase or decrease by the same percentage, assuming that the resistance is constant value.

To draw a graph of current verses voltage, let's take a constant value of resistance. For example, $R=10\Omega$, and calculate the current for several value of voltage ranged from 0 to 100 V by used the formula $I=V/10$ where $R=10\ \Omega$. the current values obtained are shown in the table (4.1):

V(v)	I(A)
10	1
20	2
30	3
40	4
50	5
60	6
70	7

$$I = \frac{V}{10\ \Omega}$$

Table (4.1)

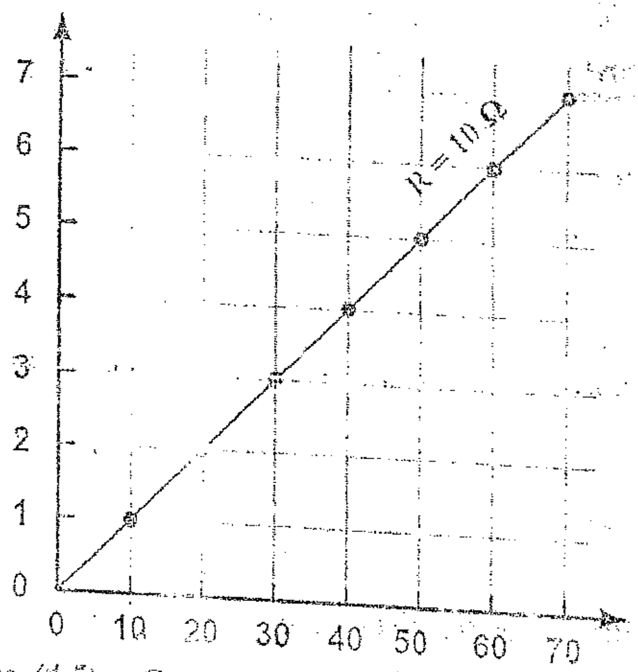


Figure (4.5) : Graph of current versus voltage for R=10 Ω

The graph of the current values versus the voltage values is shown in the Fig. (4.5). This graph tells us that a change in voltage results in a linearly proportional change in current. From graph the change in voltage from 20 to 30 is increased by 50% by calculation the current

...must be incr... by the same percentages 50%

$I = 2 + 2 \times 50/100 = 3A$.

2.4.3.2 Current and Resistance are Inversely Related

Current varies inversely with resistance as expressed ohms law, I=V/R. when the resistance is decreased, the current goes up; when the resistance is increased, the current goes down. For example, let's take a constant value of voltage V=10volts. and calculate the current for several value of resistance ranged from 10 to 100 Ω by used the formula I=10/R. the current values obtained are shown in the next table (3-2), The graph of the current values versus the voltage values is shown in the figure(3-6).

R (Ω)	I (A)
10	1
20	0.500
30	0.333
40	0.250
50	0.200
60	0.167
70	0.143

$I = \frac{10\,V}{R}$

Table (4.2)

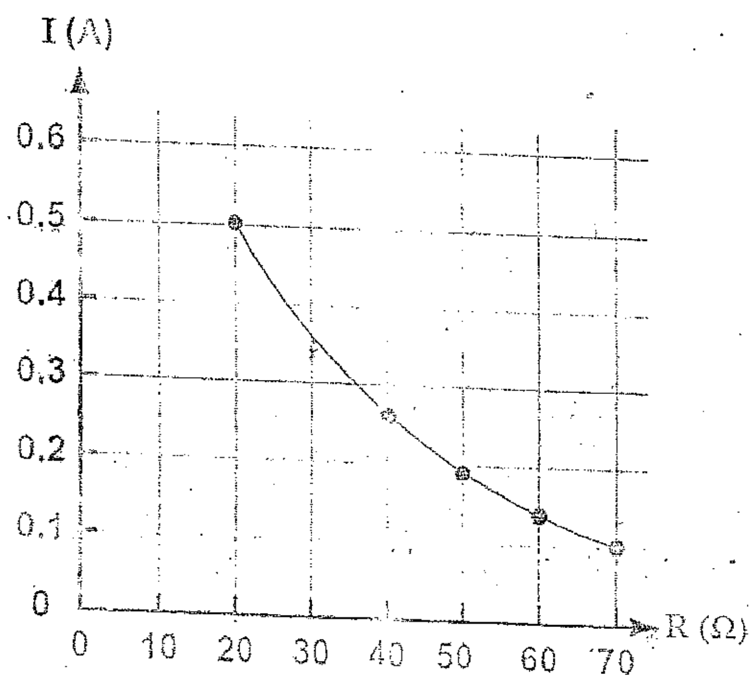


Figure (4.6) : Graph of current versus resistance for V= 10 v

2.4.4 Calculating Current

In this section you will learn to determine the current values when you known the values of voltage and resistance. As examples, by using the following formula I = V / R. In order to get current in amperes, you must express the value of voltage in volt and the value of resistance in ohms.

Example 5.1

The voltage supplied by the battery is 10 volts, and the resistance is 5 Ω ~~in the circuit of figure~~ (3-7). Calculate the current in amperes?

V
10v

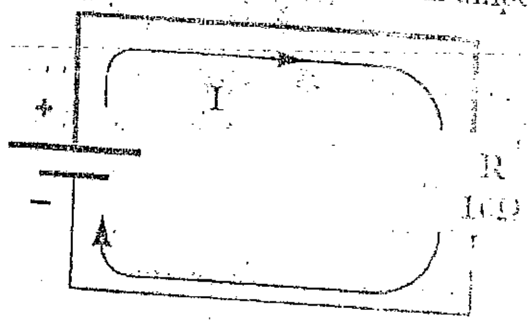


Figure (6.1)

Solution: use the formula

$$I = \frac{V}{R}$$

$$I = \frac{10}{5} = 2A$$

V = 100 volts
I = 10 Ω
I = ?

Example5-2

If the resistance in figure 5.7 is changed to 0.1 KΩ and the voltage to 50V, what is the new value of current?

Solution: $I = \frac{V}{R}$

$$I = \frac{50}{100} = 0.5\ A$$

V = 500 volts
R = 0.1 KΩ
=(0.1 X 1000 Ω
= 100 Ω
I = ?

2.4.6 Calculating Resistance

In this section, you will learn to determine the resistance values when you known the values of voltage and current. As examples, by using the following formula **R = V/I** In order to get resistance by ohms, you must express the value of voltage in volts and the value of current in amperes

Example 5.5

In the circuit in figure (3-9), how much resistance is needed to draw 4mA of current ?

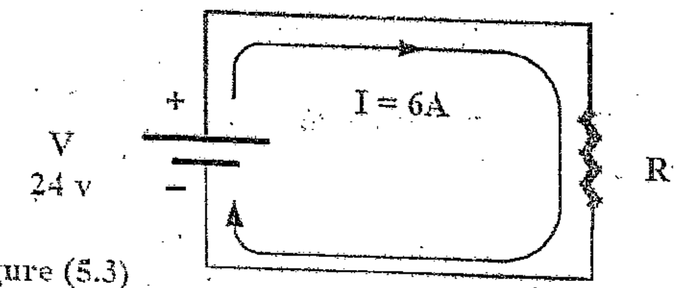


Figure (5.3)

Solution:

use the formula R= V / I

$$R = \frac{V}{I}$$

$$R = \frac{24}{6} = 4\ \Omega$$

V = 24 v

R = ?

I = 6A

Example 5.6

The circuit in figure (5.3), the voltage is changed to 1.2 KΩ, and the current is changed to 4 mA, what is the resistance

Solution:

use the formula R= V / I

$$R = \frac{V}{I} = \frac{1200}{4\times1000} = 3{,}000{,}000\ \Omega$$

If required Resistance by M Ω

R = 3,000,000 /1,000,000 = 3 MΩ

V = 1.2 kV

= 1.2 X 1000

= 1200V

R = ?

I = 4 mA

2.[illegible] [illegible]TIV[illegible]Y AND CONDUCTIVITY

Resi[illegible] ρ of a material is the resistance of unit length of material and [illegible] area at a given temperature [illegible] resistance, R, of a given materia[illegible]

- direct[illegible] proport[illegible] to the length of the material, l, and
- invers[illegible] proportional to the cross sectional area of the material, A.

mathematically, [illegible] two statements can be combined and stated as

$$R = \alpha \frac{L}{A} \tag{5.1}$$

where ρ is a constant of proportionality known as resistivity of the material, (ρ) [illegible] letter [illegible]ounce as Rho.

[illegible] equation (5.1) we can deduce that

$$\rho = \frac{R \times A}{L} \left\{ \frac{\text{Ohm} \times \text{meter}^2}{\text{meter}} = \text{Ohm-m} \right\}$$

We should note that when R is in ohms, A in m2 and l in m, then the unit of is ohm[illegible] $(\Omega - m)$

Co[illegible] The inverse of resistivity is called conductivity and is denoted by the symbol σ. (σ is a Greek letter pronounced sigma) its unit of measurement is

$$(\Omega - m) \quad \text{or} \quad \frac{1}{\Omega - m}.$$

[illegible], the two terms can be related as

$$\sigma = \frac{1}{\rho} (\Omega - m) \tag{5.2}$$

Thi[illegible] expressed as follows: the [illegible] the resistivity of a ma[illegible] conductivity, and vice versa

2.6 SERIES AND PARALLEL CIRCUITS

Series circuits A series circuit is formed when any numbers of resistors are connected end-to-start so that there is only one path for current to flow.The resistors can be actual resistors or other devices that have resistance like lamps.

Figure (6.1) shows two resistors connected in series (end -to –start) between points A and B. There is only way for current to get from point A to point B. a series circuit provides only one path for current between two points in a circuit so that the current in the same through R_1 and R_2.

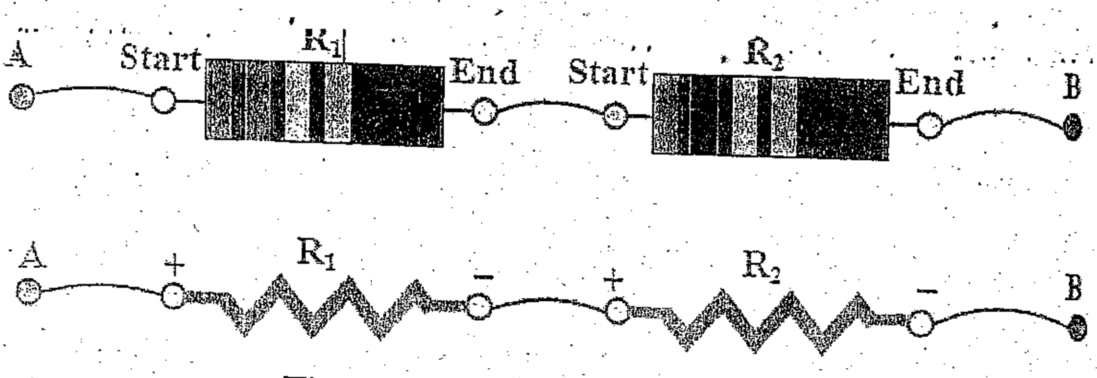


Figure (6.1) : Resistor in series

In an actual circuits diagram, a series circuit may be easy to identify as those in figure (6.1) For example. Figure (6.2) shows series resistors drawn in other ways.

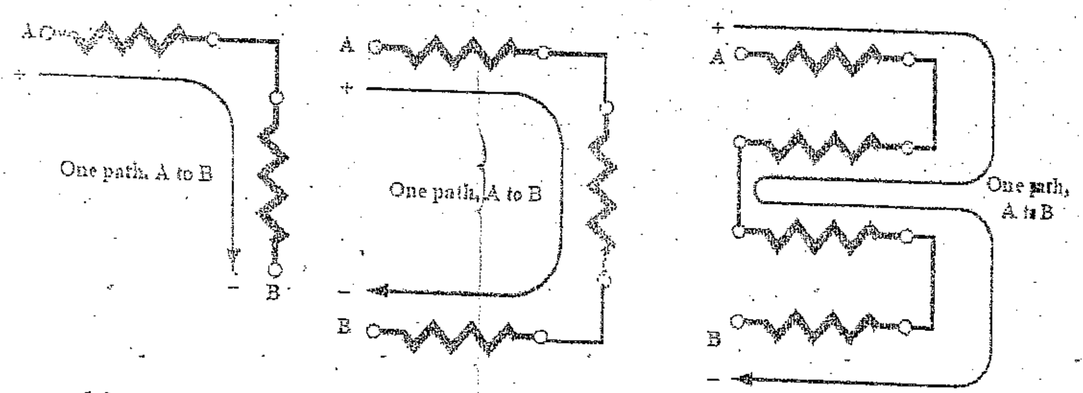


Figure (6.2) : Some examples of series circuits. Notice that the current is the same at all points

Example 7.2 A wire of diameter 0.6mm, resistivity $1.1 \times 10^{-6}\Omega$-m has resistance of 44Ω. Calculate the length of the wire.

Solution

Here, we are given:

$$\text{radius, } r = \frac{0.6}{2}\text{ mm} = 3 \times 10^{-4}\text{m}$$

To find l we deduce it from $\rho = \frac{RA}{l}$ (i.e. $l = \frac{RA}{\rho}$)

$$\therefore\ l = \frac{44 \times (^{22}/_{7}) \times (3 \times 10^{-4})^2}{1.1 \times 10^{-6}}$$

$$= 11.3\text{m}$$

Example 7.3 A 30m conductor of cross sectional area 2mm^2 has a resistance of 10 ohms. Calculate the conductivity of the conductor.

Solution

$$\rho = \frac{l}{RA}$$

$$= \frac{30}{10 \times 2(10^{-3})^2} = 1.5 \times 10^6\ (\Omega - \text{m})^{-1}$$

$$\frac{3960}{r}$$

Parallel Circuits

A parallel circuit is formed when any number of resistors are connected starts-to-starts and ends-to-ends so that there is different paths for current to flow. The resistors can be actual resistors or other devices that have resistance like lamps. Figure (6.3) shows two resistors connected in parallel (start-to-start and end-to-end) between points A and B.

There is two ways for current to get from point A to point B, a parallel circuit provides more than one path for current between two points in a circuit so that the current flow in two paths current flow in R_1 is called I_1 and current flow in R_2 is called I_2.

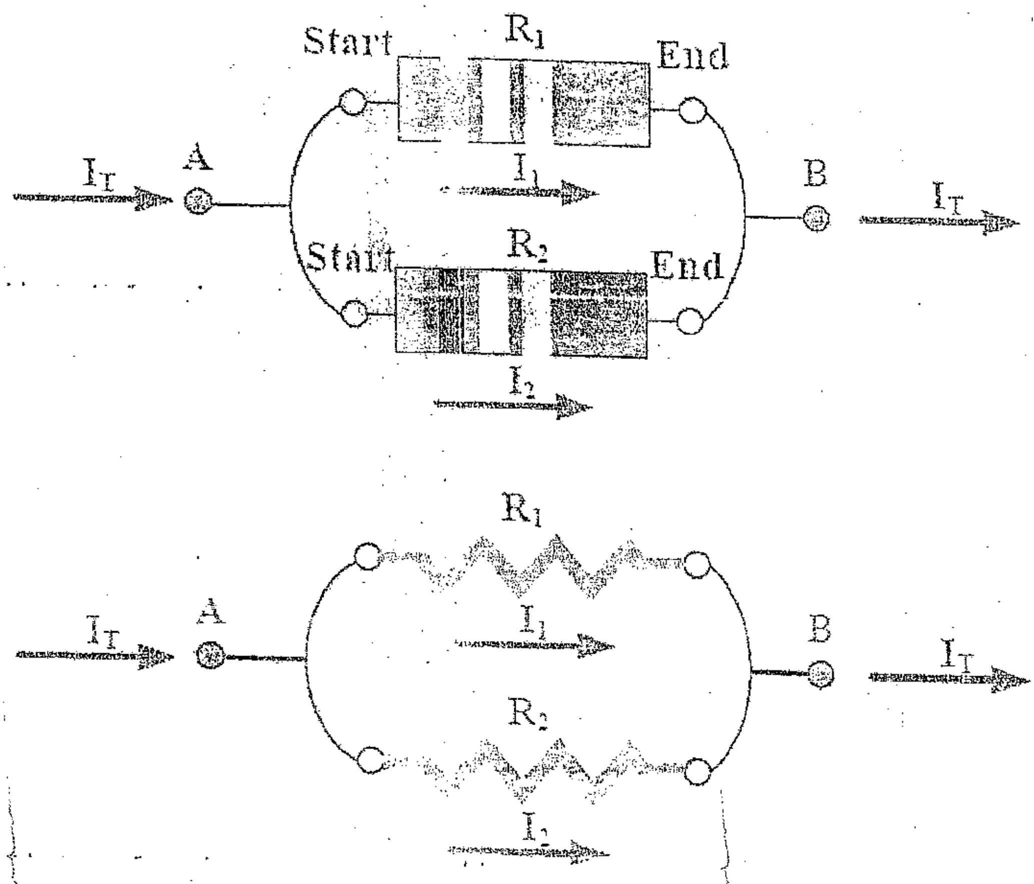


Figure (6.3) : Resistor in parallel

In an actual circuits diagram, a parallel circuit may be easy to identify as those in figure (6.3). For example. Figure (6.4) shows Some examples of parallel resistors drawn in other ways.

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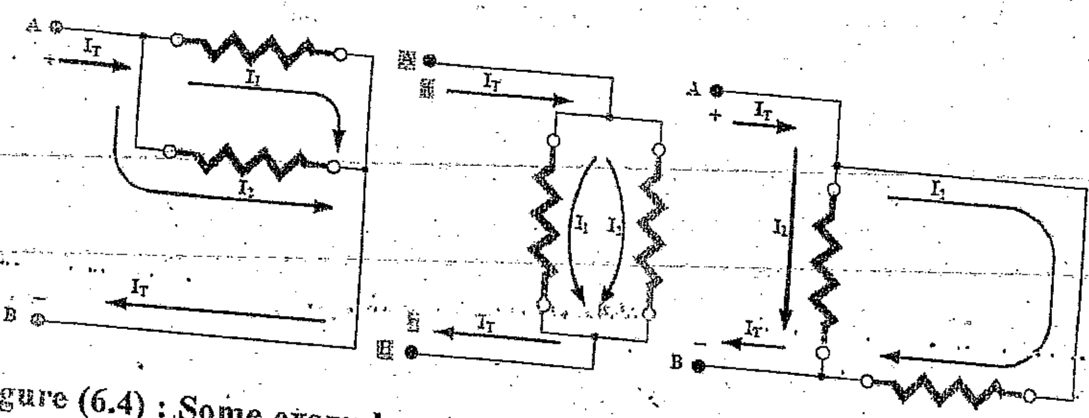


Figure (6.4) : Some examples of parallel circuits. Notice that the current is the same at all points

2.7 SOLVE PROBLEMS INVOLVING RESISTIVITY AND CONDUCTIVITY

Example 7.1 A cylinder wire of 0.5m in length, 0.5mm in diameter, has a resistance of 2.5Ω. calculate the resistivity of the wire.

Solution

length, $l = 0.5\text{m}$

radius $r = \frac{0.5\text{mm}}{2} = 0.25 \text{ mm} = 2.5 \times 10^{-4}\text{m}$

Cross sectional area, $A = \pi r^2 = \frac{22}{7} \times (2.5 \times 10^{-4})^2 \text{ m}^2$

$$\rho = \frac{RA}{l} = \frac{2.5 \times (22/7) \times (2.5 \times 10^{-4})^2}{0.5}$$

$\rho = 98.19 \times 10^{-8}$ Ohm – meter.

0.6/2

Example 7.2 A wire of diameter 0.6mm, resistivity $1.1 \text{ X } 10^{-6}\Omega\text{-m}$ has resistance of 44Ω. Calculate the length of the wire.

Solution

Here, we are given:

$$\text{radius, } r = \frac{0.6}{2}\text{ mm} = 3 \text{ X } 10^{-4}\text{m}$$

To find l we deduce it from $\rho = \frac{RA}{l}$ (i.e. $l = \frac{RA}{\rho}$)

$$l = \frac{44 \text{ X } (^{22}/_{7}) \text{ X } (3 \text{ X } 10^{-4})^2}{1.1 \text{ X } 10^{-6}}$$

$$= 11.3\text{m}$$

Example 7.3 A 30m conductor of cross sectional area 2mm^2 has a resistance of 10 ohms. Calculate the conductivity of the conductor.

Solution

$$\rho = \frac{l}{RA}$$

$$= \frac{30}{10 \text{ X } 2(10^{-3})^2} = 1.5 \text{ X } 10^{6}\,(\Omega\text{ - m})^{-1}$$

2.8 EQUIVALENT RESISTANCE OF SERIES AND PARALLEL CIRCUIT

Equivalent Resistance Of A series Circuit

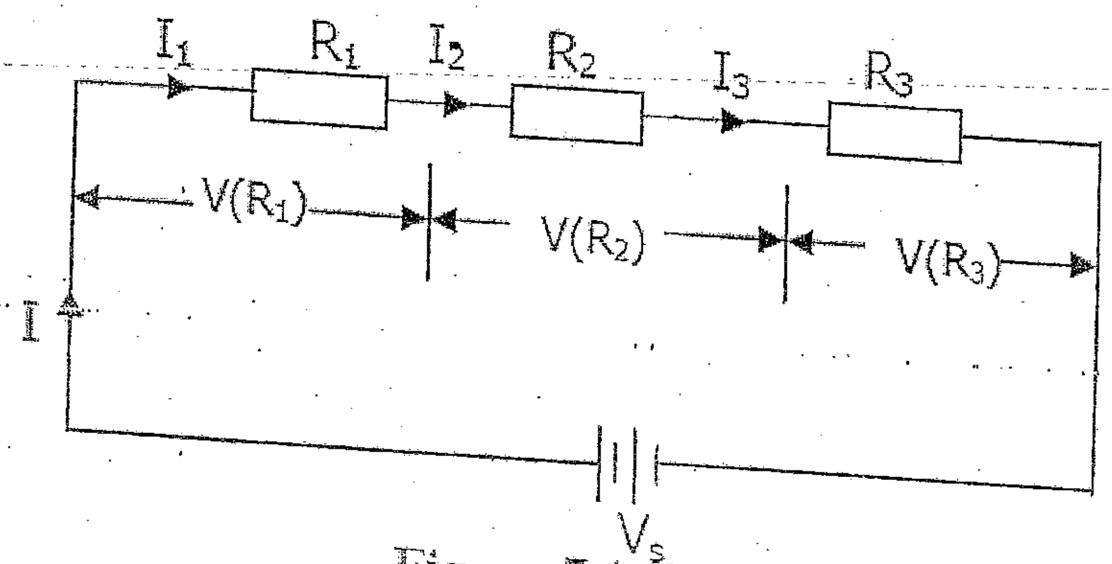


Figure 7.1 Series circuit

When resistors of resistance R_1, R_2 and R_3 are joint together (fig 7.1), they are said to be connected in series. It should be noted that:

(i) The current through a series circuit is the same all over.

(ii) The voltage drop across each resistor is different depending on its resistances.

(iii) The applied voltage (Vs) is equal to the sum of the P.d's across each resistor.

i.e. $I = I_1 = I_2 = I_3$

$$V_s = V(R_1) + V(R_2) + V(R_3) \quad (7.1)$$

where $V_s = IR$; $V(R_1) = IR_1$; $V(R_2) = IR_2$; $V(R_3) = IR_3$ (7.2)

Putting eqtn (7.2) in (7.1) gives

$$IR = IR_1 + IR_2 + IR_3$$

$$\Rightarrow Req = R_1 + R_2 + R_3$$

$V(R_2) = IR_2$

For n resistors connected in series, the equivalent resistance

Req. is given by $Req. = R_1 + R_2 + R_3 + \dots + R_n$ (7.3)

Equivalent Resistance Of A Parallel Circuit

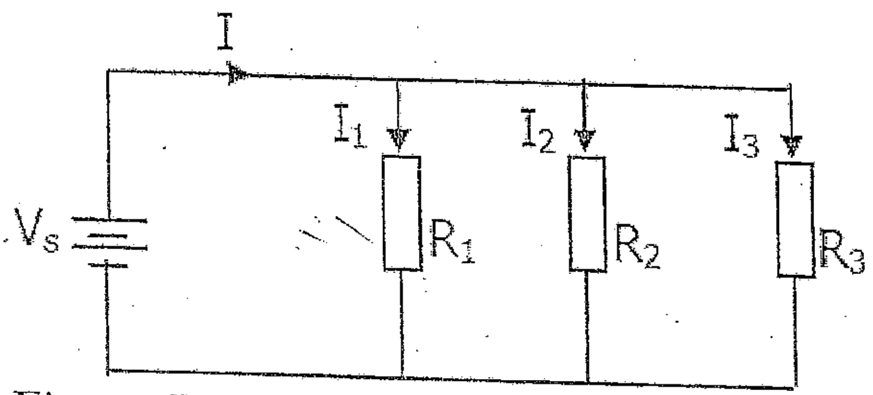


Figure 7.2 Parallel circuit

When resistors of resistance R_1, R_2 and R_3 are connected in parallel, the following statements hold good:
(i) The p.d across each resistor is the same

(ii) The current flowing through each resistor is different

(iii) The supply current (I) is equal to the sum of current flowing through each resist

i.e. $Vs = V_1 = V_2 = V_3$

$I = I_1 + I_2 + I_3$ (7.4)

where $I = Vs/R$; $I_1 = Vs/R_1$: $I_2 = Vs/R_2$; $I_3 = Vs/R_3$ (7.5)

Inserting eqtn (7.5) in (7.4) gives

$$Vs/R = Vs/R_1 + Vs/R_2 + Vs/R_3$$

$$\Rightarrow 1/R = 1/Req = 1/R_1 + 1/R_2 + 1/R_3$$

For n resistors connected in parallel,

$$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3 + \text{- - -} + 1/R_n$$

Example 7.4
In the circuit in figure (7.3), Determine the two voltages drops across R1 and R2?

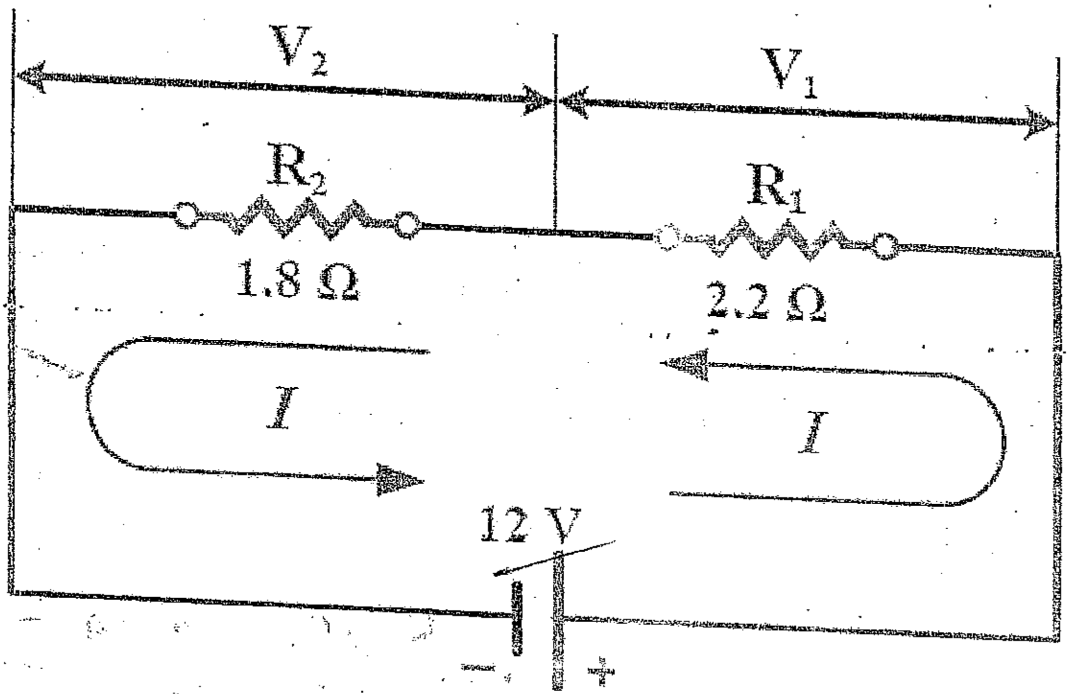


Figure 7.3 $\frac{12v}{R_T} = \frac{12}{4} = \underline{6\ Am}$ ie $I_T = 6$

R_2 R_1 $1 = 1_2 = 1_3$

1.8 2.2 $V_T = V_{dR_1} + V_{dR_2} = 6.\cancel{8}\times 5.4$ $= 3$

$= 1.00$

$1.8 + 2.2 = 4$

$12v$

$V_{R_1} = I \times R_1 = 3 \times 1.8 = \cancel{\frac{12v}{1.8}} = 6.67$

3×2.2 ~~Vd~~ across R_1

$\Rightarrow V_1 = I_T \times R_1 = 3 \times 1.8 = 5.$

Solution:

1) First solve for total resistance.	$R_1 = 2.2\ \Omega$
$R_T = R_1 + R_2$	$R_2 = 1.8\ \Omega$
$= 2.2 + 1.8 = 4\ \Omega$	$V = 12\ v$
2) Second, solve for current:	$I = ?$
$I = \dfrac{V}{R_T} = \dfrac{12}{4} = 3\ \text{Amp}$	$V_{R1} = ?$
3) Finally, solve for voltage drop across any resistor:	$V_{R2} = ?$

<u>Voltage drop across R_1</u>

$V_{R1} = I \times R_1 = 3 \times 2.2 = 6.6\ v$

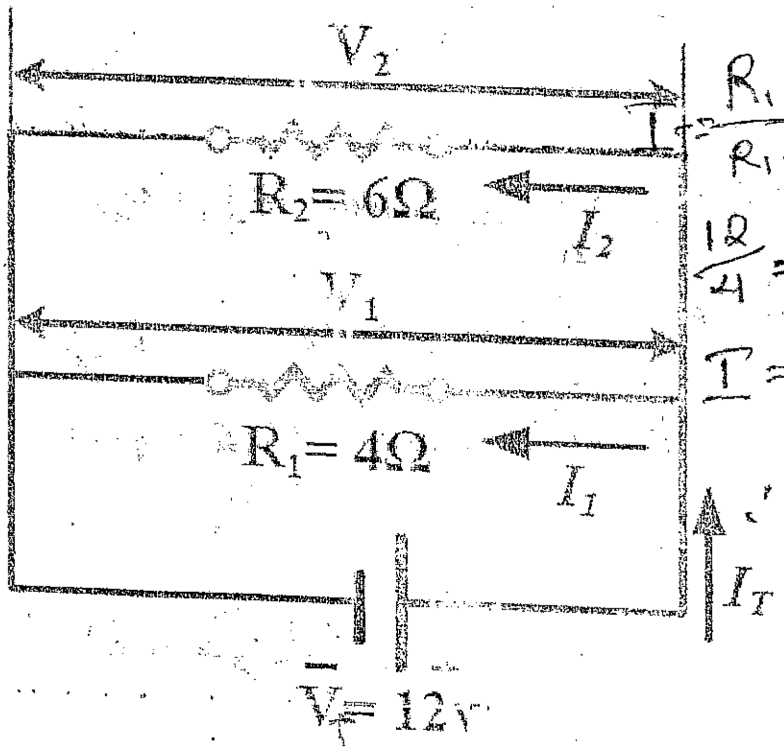
<u>Voltage drop across R_2</u>

$V_{R2} = I \times R_2 = 3 \times 1.8 = 5.4\ v$

The sum of all voltage drops (6.6v + 5.4v) equals the source voltage (12v).

Example 7.5

In the circuit in figure (7.4), Determine the current in each branch



$R_T = \dfrac{R_1 \times R_2}{R_1 + R_2} \quad \dfrac{6 \times 4}{6+4} = \dfrac{24}{10} = 2.4$

$\dfrac{12}{4} = I = \dfrac{V}{R} = 3A \cdot R_1$

$I = \dfrac{V_2}{R_2} = \dfrac{12}{3} = 4A$

$\therefore I_T = 3 + 4 = 7A$

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Figure 7.4

Solution:

1) First solve for total resistance.

Used the formula when there are two unequal value resistors

$$R_T = \frac{R_1 X R_2}{R_1 + R_2} = \frac{4 X 6}{4 + 6} = \frac{24}{10} = 2.4\ \Omega$$

2) Second, solve for voltage across any resistor:

$V_{R1} = V_{R1} = V_T = 12$ v

3) Finally, solve for current in each branch :

$$I_1 = \frac{V_{R1}}{R_1} = \frac{12}{4} = 3 \text{ Amp}$$

$$I_2 = \frac{V_{R1}}{R_1} = \frac{12}{6} = 2 \text{ Amp}$$

$$I_T = \frac{V_T}{R_T} = \frac{12}{2.4} = 5 \text{ Amp}$$

$R_1 = 4\ \Omega$
$R_2 = 6\ \Omega$
$V = 12$ v
$I_1 = ?$
$I_2 = ?$

The sum of current in each branch (3 + 2) equal the total current 5A

2.10 KIRCHHOFF'S LAWS

Kirchhoff's laws were developed in 1847, by the German physicist G.R. Kirchhoff. These laws are two in numbers and they are formally known as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL). We get.

a node is KCL states that the sum of I entering the same sum of I leaving the node.

Kirchhoff's current law

$$i_1 + (-i_2) + (-i_3) + i_4 + i_5 = 0 \qquad (8.1)$$

or $$i_1 + i_4 + i_5 = i_2 + i_3 \qquad (8.2)$$

Definition of KCL

From equation (8.2) we notice that the sum of current entering a node is equal to the sum of the current leaving the node. This is an alternative way of stating KCL

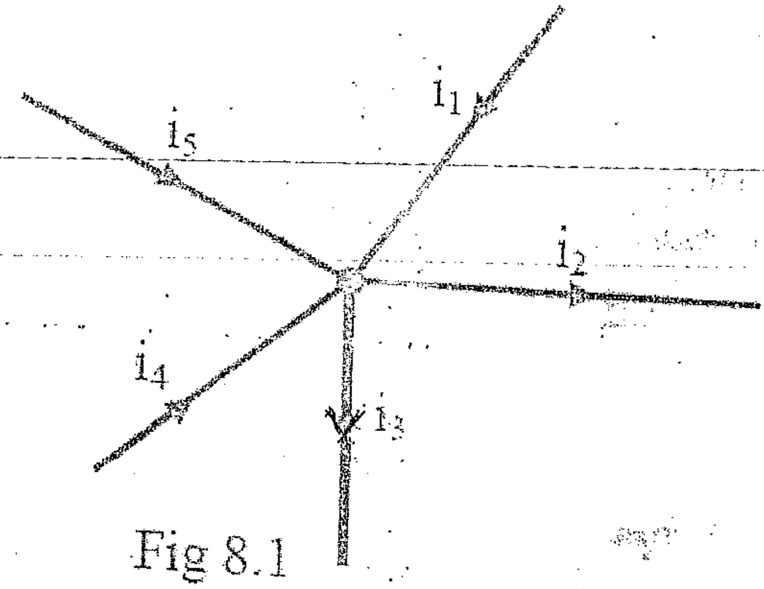
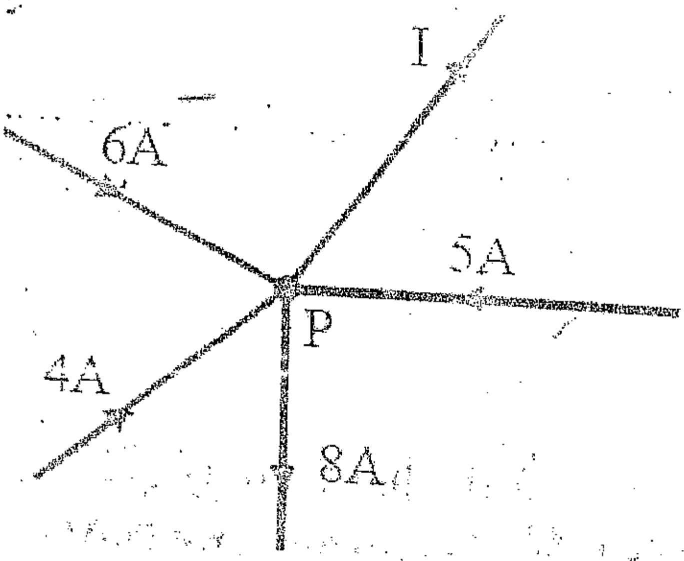


Fig 8.1

Example 8.1
Find the value of I in Fig 8.2



Solution

In order to find the value of I, the following equation can be formed. Currents towards P = currents flowing from P.

Therefor $6 + 4 + 5 = I + 8, \Rightarrow I = 7A$

KVL states that the algebric Sum of the Emf in the Circuit is the Same as algebric Sum of Voltage drops. $\pm E_1 + \pm E_2 = \pm I_1R_1 + \pm I_2R_2$

Kirchhoff's voltage law

Kirchhoff's voltage law state that in a closed circuit, the algebraic sum of the e.m.f.s is equal to the algebraic sum of the voltage drops.

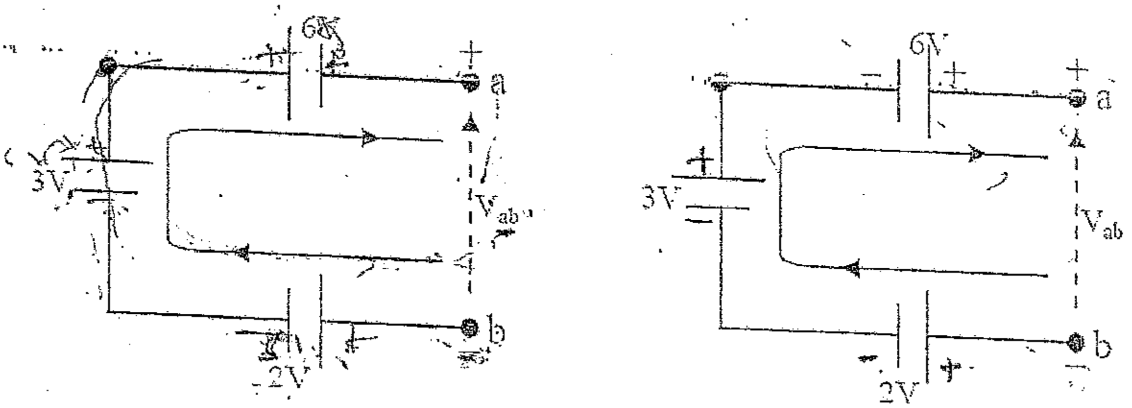
Mathematically this can be stated as: $\pm E_1 + \pm E_2 + = \pm I_1R_1 + \pm I_2R_2 +$

$$\pm E_1 + \pm E_2 = \pm I_1R_1 \pm I_2R_2 \cancel{\pm I_3R_3}$$

if there are two sources of e.m.f and three resistors R1, R2 and R3

In general, $\sum E = \sum IR$

Example 8.2 Determine the value of V_{ab} in the circuit shown in Fig 8.3 (a) and (b) using KVL.



(a) (b)

Fig 8.3

Solution

For Fig 8.3(a), applying KVL, we get

$$-2V + 3V - 6V - V_{ab} = 0$$

$$\therefore \quad V_{ab} = -5V$$

For Fig 8.3 (b) applying KVL, we get

$$-2V + 3V + 6V - V_{ab} = 0$$

$\therefore \quad V_{ab} = 7V$

Example 8.3 Applying KCL and KVL to the circuit of Fig. 8.4, write down the equation pertaining to loop I and II respectively,

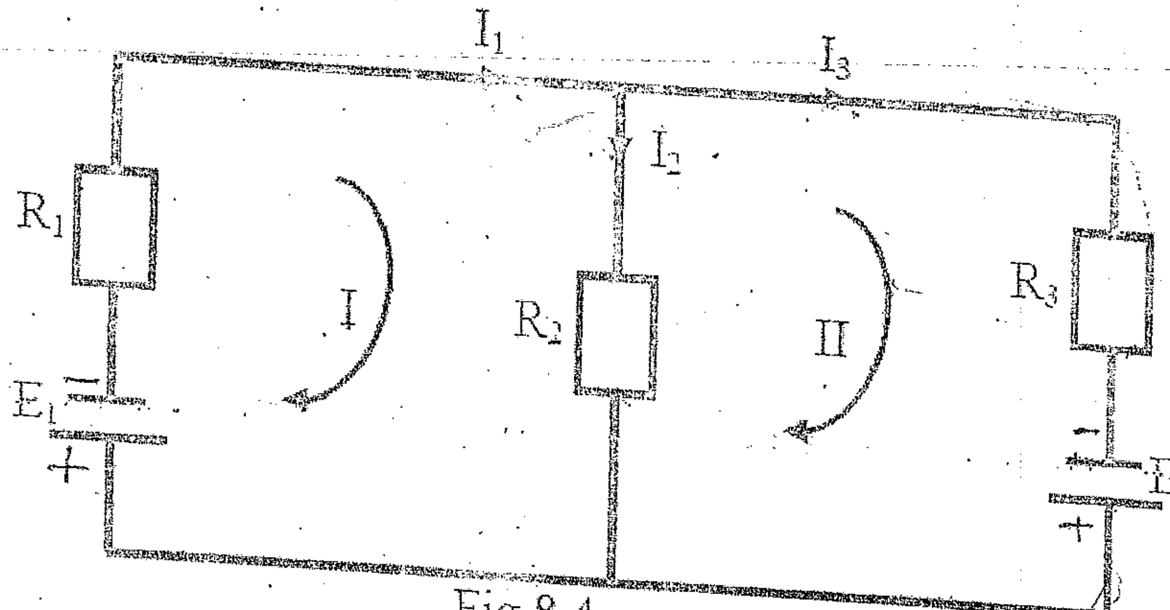


Fig 8.4

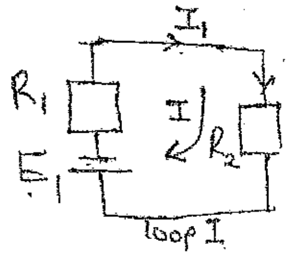
Solution

N.B. we are working under the assumption of fixed direction of I1, I2 and I3 as shown in Fig 8.4

Applying KVL to loop I, we get

$$-E_1 = R_1I_1 + R_2I_2$$

CL

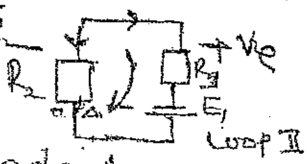


Consider
-ve sign fr
$-E_1$

Applying KVL to loop II, we get

$$E_2 = -R_2I_2 + R_3I_3$$

-ve



From KCL we known that at node A

$$I_1 = I_2 + I_3$$

I -ve sign	II +ve sign
CL Direction	ACL - Direction
-ve $-E_1 = R_1I_1 + R_2I_2$	$+E_2 = -R_2I_2 + R_3I_3$

or $\quad I_3 = I_1 - I_2$

Putting this value of I_3 into the equation involving E_2, we get

$$E_2 = -R_2I_2 + R_3\,(I_1 - I_2)$$

$$E_2 = R_2I_2 - (R_2 + R_3)\,I_2$$

2.11 SUPERPOSITION PRINCIPLE

The superposition principle is a method that allows us to determine the current through or the voltage across any resistor or branch in a network. The advantage of using this approach instead of Kirchhoff's laws is that it is not necessary to use determinant or matrix algebra to analyze a given circuit. The theorem state the following:

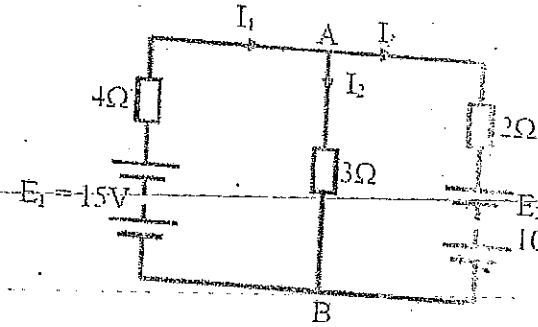
In a circuit containing several sources of e.m.f., the resultant current in any branch is the algebraic sum of the currents in that branch that would be produced by each e.m.f, acting alone, all other sources of e.m.f being replaced meanwhile by their respective interval resistances.

In order to apply the superposition principle it is necessary to remove all sources other than the one being examined. In order to ''Zero'' a voltage source, we **replace** it with a short **circuits**, since the voltage across a short circuit is zero volts. A current source is zeroed by **replacing** it with an open **circuit**, since the current through an open circuit is zero amps.

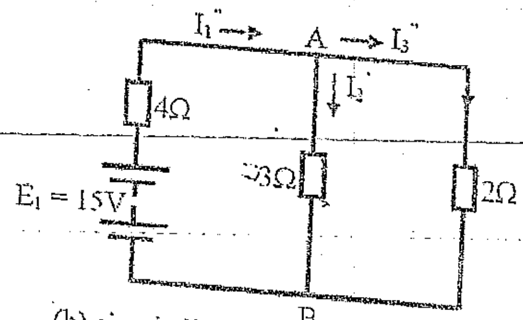
Example 8.1 Solve the same problem in example 3.5 to determine all the branch currents using the superposition principles.

Solution

To solve this problems we shall follow the steps. **Step** *I*, Re-draw the given circuit diagram with E1 = 15V and E_2 = OV, as shown in Fig 9.1 below



(a) Original circuit diagram



(b) circuit diagram with $E_2 = 0V$

Fig 9.1

Step *II*: Next, determine all the branch currents I_1'. I_2' and I_3' indicated in Fig 9.1 (b),

With $E_1 = 15V$ and $E_2 = 0V$.

Step *III*: we find the equivalent resistance of 3Ω resistor and 2Ω resistor connected in parallel.

i.e. $R_{3\Omega}//R_{2\Omega} = \frac{3 X 2}{3+2} = \frac{6}{5} = 1.2\Omega$

Step *IV*: The circuit diagram of Fig 9.1(b) reduces to the form shown in Fig 9.1

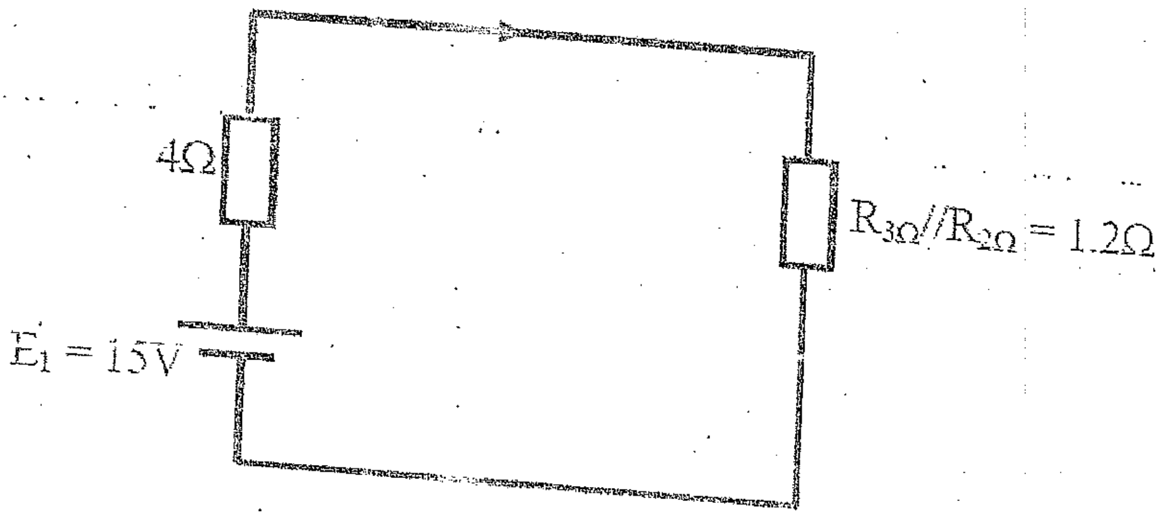


Fig 9.2

Step V: we can apply the Kirchhoff's Voltage law to the circuit diagram of Fig 9.2 to obtain,

$$I'_1 = \frac{E_1}{(4 + R_{3\Omega}//R_{2\Omega})}$$

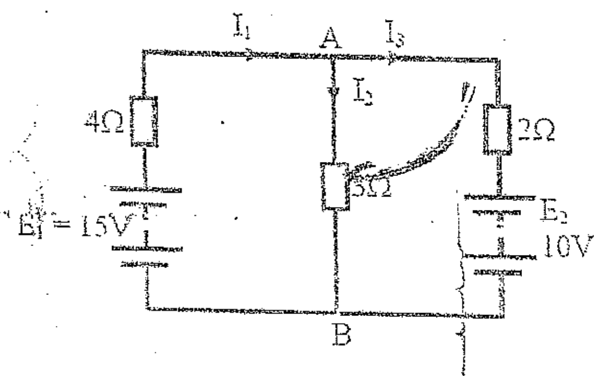
$$I'_1 = \frac{15}{(4 + 1.2)} = 2.885A$$

Step VI: Turn back to fig 9.1(b) and bear in mind that I1 is the total current that flows into the 3Ω resistor branch and 2Ω resistor branch which are in parallel. Applying the principle of current division rule, we get

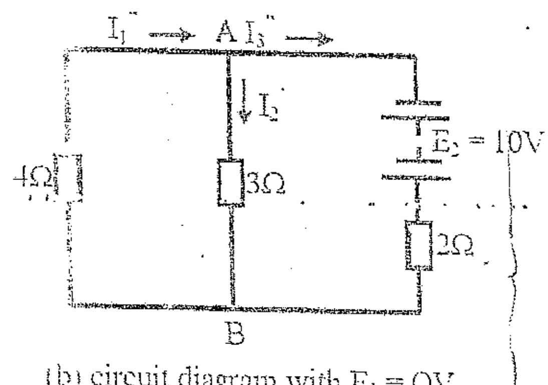
$$I'_2 = \frac{R_{2\Omega}}{(R_{2\Omega}) + R_{3\Omega})} \text{ X } I'_1 = \frac{2}{(2+3)} \text{ X } 2.885 = 1.154A$$

$$\text{Similarly, } I'_3 = \frac{R_{3\Omega}}{(R_{2\Omega} + R_{3\Omega})} \text{ X } I'_1 = \frac{3}{(2+3)} \text{ X } 2.885 = 1.731A$$

Step VII: Draw the corresponding circuit diagram if the original circuit diagram of Fig 9.1 (a) has E_1 =1OV and E_2 = 10V, to produce the circuit diagram shown in Fig. 9.3(b)



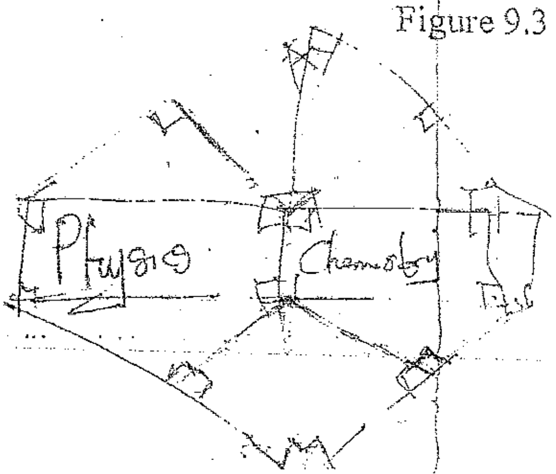
(a) Original circuit diagram



(b) circuit diagram with E_1 = OV

Figure 9.3

1Ω - 2Ω
= 0Ω



Following all the stages in step II above, we obtain the following results:

(i). $R_{4\Omega}//R_{3\Omega} = \frac{4 X 3}{4 + 3} = \frac{12}{7}\Omega = 1.714\Omega$

(ii) The resulting equivalent circuit diagram after combing $R_{4\Omega}$ and $R_{3\Omega}$ in parallel gives Fig 9.4.

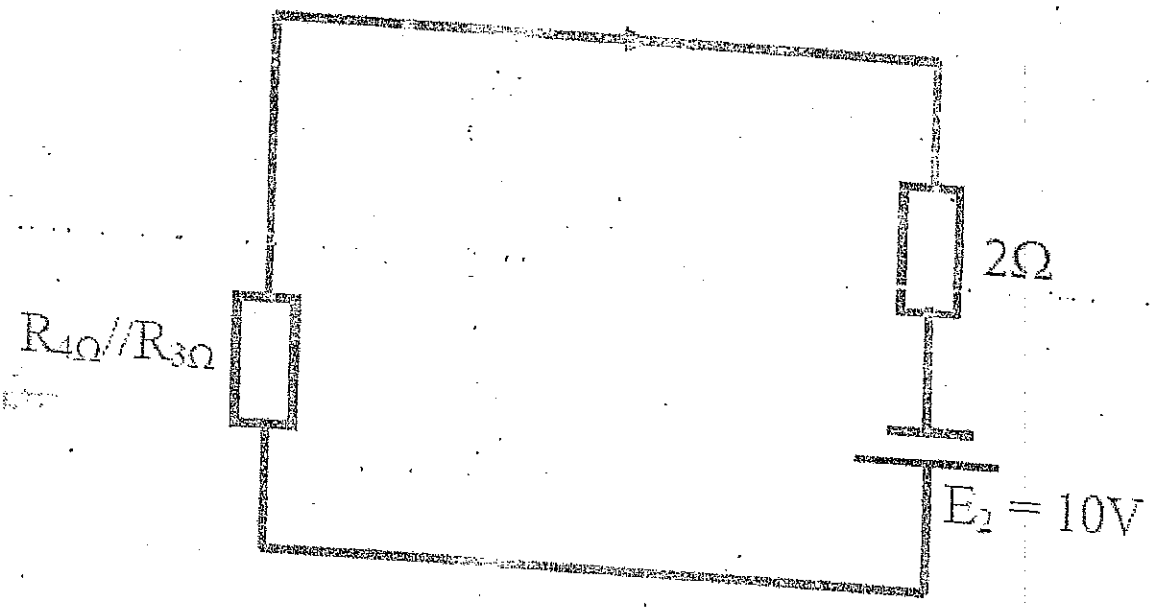


Fig 9.4

(iii) Applying Ohm's law to the circuit diagram of Fig 9.4, we get

$$I_3 = \frac{E_2}{2 + R_{4\Omega} // R_{3\Omega}} = \frac{10}{(2 + 1.714)} \doteq 2.693A$$

(iv) with reference to Fig 9.3, we see that if I''_3 is to serve as the total current then according to KCL then I''_1 and I''_2 must be seen to flow into node A. For

(v) this to happen the direction I''_2 must be opposite to what is indicated in the diagram of Fig 9.3. This means that I''_2 must bear a negative value. Only I''_1 flows into node A.

Therefore, $I''_1 = \frac{3}{(4+3)} \times 2.693 = 1.154A$

Similarly, $I''_2 = -\frac{4}{(4+3)} \times 2.693 = -1.539A$

Step VIII: The total branch currents can be obtained by adding up algebraic as follows:

$I_1 = I'_1 + I''_1 = 2.885 + 1.154 = 4.039A$

$I_2 = I'_2 + I''_2 = 1.154A - 1.538A = -0.384A$

$I_3 = I'_3 + I''_3 = 1.731A + 2.693A = 4.424A$

2.12 TEMPERATURE COEFFICIENT OF RESISTANCE

If the resistance of a coil of insulated copper wire is measured at various temperatures up to say 200^0c, it is found to vary as shown in the figure 10.1 below.

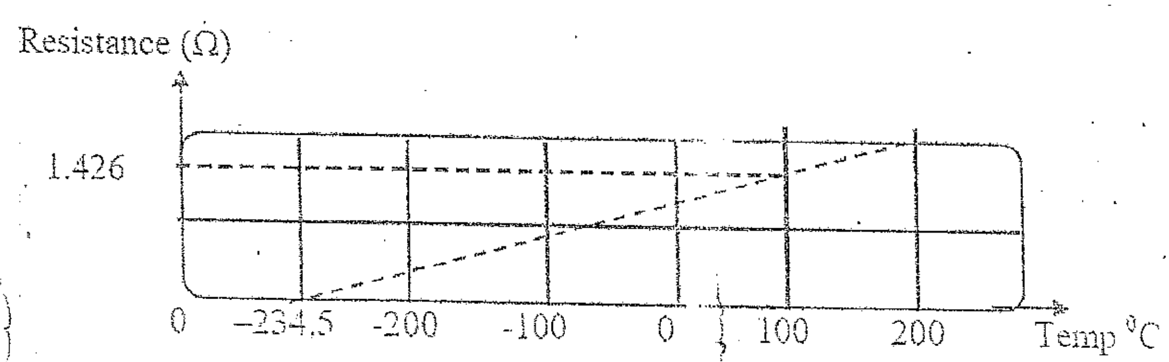


Fig 10.1 Variation of resistance of copper with temperature

The resistance at 0^0C been taking as 1Ω. The resistance increases until it reaches 1.426Ω for an increase of 100^0C in the temperature.

We may define temperature coefficient of resistance as "the increase in resistance per original resistance at 0^0C per temperature change"

Mathematically $\alpha = \dfrac{R_1 - R_o}{R_o(\theta_1 - \theta_o)}$ (10.1)

The unit of temperature coefficient of resistance is $^0C^{-1}$

Graph of resistance against temperature

The graph of resistance against temperature is shown in Fig 10.2. below.

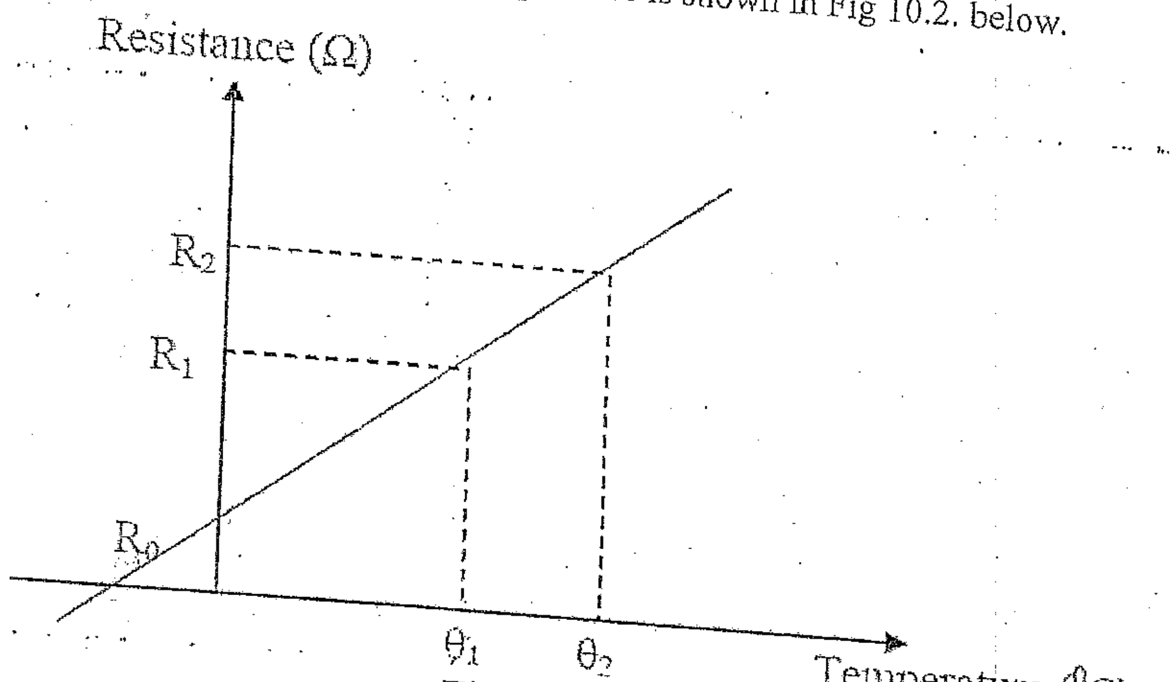


Fig 10.2

Slope $= \dfrac{R_2 - R_1}{\theta_2 - \theta_1}$, also slope $= \dfrac{R_1 - R_o}{\theta_1 - \theta_o}$

Temperature coefficient of resistance $= \dfrac{\text{slope of resistance/ temperature}}{\text{Resistance}}$

i.e. $\alpha_1 = \dfrac{R_2 - R_1}{R_1(\theta_2 - \theta_1)}$ (10.2)

$\alpha_0 = \dfrac{R_1 - R_2}{Ro(\theta_1 - \theta o)}$ (10.3)

From (10.3), $\theta o = 0$

$\Rightarrow \alpha_0 = \dfrac{R_1 - R_2}{R_0\,\theta_1}$

$=> \alpha_0 = R_o (1 + \alpha \theta_1)$, where $\alpha_0 = \alpha$ (10.4)

Also, $R_2 = R_0(1+\alpha \theta_2)$ (10.5)

Generally $\boxed{R_\theta = R_0 (1+\alpha \theta)}$ (10.6)

From (10.4) and (10.5)

$$\frac{R1}{R2} = \frac{R_0 (1+\alpha \theta 1)}{R_0 (1+\alpha \theta 2)}, => \boxed{\frac{R_1}{R_2} = \frac{1+a \theta_1}{1+a \theta_2}} \quad (10.7)$$

where R_1=Resistance at temperature θ_1

R_2= " " " θ_2

α = Temperature coefficient of resistance

Example 10.1 A coil has a resistance of 50Ω at 0^0C, if the temperature coefficient of resistance for the coil is 0.0043Ω/Ω^{0}C, determine the resistance of the coil at 25^0C.

Solution

$$R_\theta = Ro\ (1+\alpha\theta) = 50(1+0.0043X25) = 55.38\Omega$$

Example 10.2 The resistance of a coil of a copper wire at the beginning of a heat test is 173Ω, the temperature being 16^0C. At end of the test, the resistance is 212Ω. Calculate the temperature raise of the coil. Assuming the temperature coefficient of resistance of copper to be 0.00426/^{0}C at 0^0C.

Solution

$$R_1 = 173\Omega, R_2 = 212\Omega, \theta_1 = 16^0C,\ \theta_2 = ?, \alpha = 0.00426/^0C$$

$$\frac{R_1}{R_2} = \frac{1+\alpha\theta_1}{1+\alpha\theta_2}$$

$$\theta_2 = \frac{0.30896}{0.00426} = 72.52^0C$$

Raise in temperature = $\theta_2 - \theta_1 = 72.5 - 16 = 56.5^0C$

Example 10.3 A copper wire has a resistance of 85Ω at 15⁰C. Calculate its resistance at 90⁰C. Assuming the temperature coefficient resistance of copper to be 0.0043/⁰C,

Solution

$Q_1 = 15^0C$, $Q_2 = 90^0$, $R_1 = 85\Omega$ and $\alpha_0 = 0.00\text{-}43/^0C$,

$$\frac{R_{15}}{R_{90}} = \frac{1 + \alpha_0 \theta_1}{1 + \alpha_0 \theta_2} = \left(\frac{1 + 0.0043 \text{ X } 15}{1 + 0.0043 \text{ X } 90}\right)$$

i.e. $$R_{90} = R_1 \left(\frac{1 + 0.0043 \text{ X } 90}{1 + 0.0043 \text{ X } 15}\right)$$

$\therefore$ $$R_{90} = 110.75\Omega$$

3.1 ENERGY AND ITS VARIOUS TYPES

Energy

Energy is the ability to do work. The SI unit of energy is the Joules. The Joules can be defined as the work done when a force F of 1N (Newton) acts through a distance d of 1m in the direction of the force.

Work done = F (Newton) x d (meters)

There are many forms of energy; namely;

Electrical Energy

Mechanical Energy

Thermal Energy

Light Energy

Chemical Energy

Wind Energy

Solar Energy

Atomic Energy, e.t.c.

3.2 RELATIONSHIP BETWEEN ELECTRICAL, MECHANICAL AND THERMAL ENERGY

One form of energy can be converted into another by means of a suitable process. For example, an electric motor converts electric energy into mechanical energy. The various forms of energies are related by the law of conversation of energy.

The Law of Conversation of Energy states that Energy can neither be created nor destroyed. This means that when needs to be converted; the total energy in a system is equal to the total energy output from the system. For example a car in motion has mechanical (Kinetic) energy and when the brake are applied for it to come to rest, this mechanical energy is converted into heat energy and sound energy.

Consider an electric fan. For an electric fan to rotate it takes voltage supplies from the main supply and causes the electric motor to rotate. This action in turn causes the blades of the fan to rotate and produce wind energy (a form of mechanical energy) and sound energy. In this case, electric energy was converted to mechanical energy and sound energy.

3.3 ELECTRICAL POWER

Power is the rate of doing work. The unit of Power is *joules/second or watt.* Lager units of power could be *kilowatt*. Also in dealing with large amount of energy, it is sometimes more convenient to express energy in *kilowatt hour* than in *joules*.

1*KWh*=1000 *watt hours*

=1000 x 3600 *watt seconds* or *joules*

=3,600,000J = 3.6 *Mega joules*

From its definition the expression for Electric Power can be written as;

Power = work done /time taken

Therefore Electric Power can be written as;

$$P = IV, \text{ joules}$$

Or $$P = I^2R, \text{ joules}$$

Or $$P = \frac{V^2}{R}, \text{ joules}$$

3.4 JOULE'S LAW

Electrical Energy *E* is the energy required for *V* volts of electricity to make an electric current of *I* amperes to flow through a conductor of resistance *R* Ohms for a time *T* seconds

$E = IVt$, joules

Or $$E = I^2Rt, \text{ joules}$$

Or $$E = \frac{V^2}{R}t, \text{ joules}$$

The above expressions are also known as Joules Law, which states that;

The amount of work required to maintain a current of I amperes through a resistance of R Ohms, for time t seconds is; I^2R

Example 12.1 Calculate the work done in moving through a distance of 25m by a force of 20N which acts in the direction of the motion of the body.

Solution

Since work done = F X d.

Work done = 20 x 25 = 500J

Example 12.2 A force of 40N IS applied to a body to move it at a uniform velocity through a distance of 15m in 10s in the direction of the force. Calculate the power produced.

F = 40N $P = \frac{wd}{T}$ wd = F x d 600

D = 15m

T = 10s ∴ $P = \frac{600}{10}$ = 60 watt